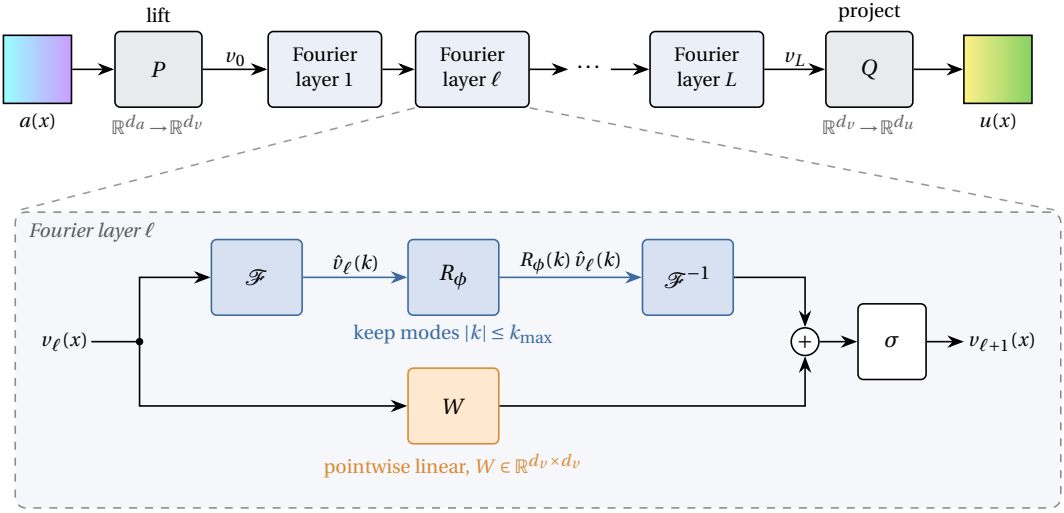

$$v_{\ell+1}(x) = \sigma\Big(W v_\ell(x) + \mathscr{F}^{-1}\big(R_\phi \cdot \mathscr{F} v_\ell\big)(x)\Big)$$